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A novel hybrid inception-xception convolutional neural network for efficient plant disease classification and detection

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Plants are essential at all stages of living things. Plant pests, diseases, and symptoms are most regularly visible in plant leaves and fruits and sometimes within the roots. Yet, their diagnosis by experts in the laboratory is expensive, tedious, and time-consuming if the samples involve laboratory analysis. Failure to detect early plant symptoms and diseases is the core biotic cause of increased plant stresses, structure, health, reduced subsistence farming, and threats to global food security. To mitigate these problems at a social, economic, and environmental level, inappropriate herbicide application reduction and early plant disease detection and classification (PDDC) are significant solutions in this case. Advancements in transfer learning techniques have resulted in effective results in smart farming and have become extensively used in disease identification and classification research studies. This study presents a novel hybrid inception-xception (IX) using a convolution neural network (CNN). The presented model combines inception and depth-separable convolution layers to capture multiple-scale features while reducing model complexity and overfitting. In contrast to ordinary CNN architectures, it extends the network for better feature extraction, improving PDDC performance that demands diverse feature competencies. It further presents a real-time artificial intelligence (AI) application available in MATLAB, Android, and Servlet to automatically identify and classify diseases based on the leaf environment using improved CNN, machine learning (ML), and computer vision techniques. To assess the presented IX-CNN model performance, different classifiers, namely, support vector machine (SVM), decision tree (DT) and random forest (RF), were used. The experiments used six datasets, including PlantVillage, Turkey Disease, Plant Doc, Rice Disease, RoCole, and NLB datasets. Plant Doc, PlantVillage, and Turkey Disease datasets demonstrated an accuracy of 100%. Rice Disease, RoCole, and NLB attained an accuracy of 99.79%, 99.95%, and 98.64%, respectively.

Keywords Zero hunger, Climate action, Computer vision, Machine learning, Plant disease detection, Good health and well-being, Plant disease classification, Life on land

Sustainable food production is paramount to global food security, and how plant diseases, symptoms, and pests can be timely detected and appropriately prevented is a trending paradigm in smart farming, such as managing the major arthropod pests affecting cassava production in Sub-Saharan Africa¹. Different human societies have been capable of producing food to adequately cater to the current and growing population using advanced technology in the agricultural sector. However, these technologies and current technologies are limited since plant diseases are caused depending on seasonal or environmental factors, such as nematodes, viruses, phytoplasmas, bacteria, fungal-like organisms, fungi, protozoa, parasitic flowering plants, and others, illustrating how the impact of *Bacillus subtilis* and *Pseudomonas fluorescens* beneficial bacterial agents on soil-borne diseases².

Calamities, for instance, plant pests, pathogens, and diseases, boost biotic and abiotic stresses, causing poor yields and even death during the completely growing life cycle³. Plant diseases are the constant disturbance caused by casual agents leading to irregular physiological processes interrupting the plant's normal function, growth, structure, or additional activities. These interferences with one or more of a crop's vital biochemical or physiological systems elicit specific pathological symptoms or conditions⁴. Crop diseases can be categorised

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based on their causal agent nature, either noninfectious or infectious. Hostile growing environments, like temperature extremes, destructive relationships between oxygen and moisture, atmosphere or soil, toxic substances, and deficiency or excess essential minerals, instigate noninfectious crop diseases⁵. Pathogenic organisms trigger infectious crop diseases. In a natural setup, crops may be pretentious by several disease-causing agents simultaneously.

Traditional farmers in underdeveloped and developing nations use their naked eyes or experience to detect and classify plant diseases and pests affecting their plants. Since plant diseases often have different causing agents, traditional farmers cannot accurately distinguish them even if they use their experience. The plant's morphology can be changed due to several factors that peasant farmers cannot differentiate⁶. Population globally by 2050 is anticipated to be 9 billion, and agricultural production worldwide should increase by at least 70% to feed the population as per the United Nations' Sustainable Development Goals (SDGs), thus contributing towards "Zero Hunger (<https://sdgs.un.org/goals/goal2>)" and "Live on land (<https://sdgs.un.org/goals/goal15>)"⁷. This is complex due to these diseases' development and transmission mechanisms. Understanding how plant disease anomalies are caused by disease development and transmission increases proper plant disease detection and classification⁸. Thus, proper use of herbicides like pesticides and insecticides reduces the application of unwanted chemicals in the soil that cause pollution. As a result, deploying artificial technologies is essential for optimising resource utilisation and raising agricultural production⁹.

Since different plant diseases have more than one different development cycle and transmission, this research focuses on the disease related to pathogenesis (this is the disease stage where the pathogen is within the already existing host tissues) in three main stages, namely inoculation, incubation, and infection, but not *saprogenesis* (the pathogen's life cycle section (part) when it is not in vigorous connotation with living host tissues). Under saprogenesis, there is constant growth in dead host tissue, which sometimes converts into dormancy¹⁰. Several environmental factors influence plant disease development, making it epiphytotic or not. Mainly include soil fertility (lowering or raising certain nutrient elements levels like pear and apple fire and Botrytis blight, septoria), soil type (some pathogens are ideal in clay and loam¹¹. For example, in cotton, there is *Phymatotrichum* root rot), soil pH (alkalinity or acidity distinctly stimulates a few diseases, such as *Plasmidiophora brassicae*), soil moisture (low or high influences the development of some root rot diseases), relative humidity (so perilous with fungal spore germination) and temperature (every pathogen has its optimal temperature intended for growth)¹². It is possible to eliminate the pathogen spread at the farm or prevent it from reaching the host if detected, classified timely, and treated appropriately¹³. This research focuses on the early detection of these pathogens to attain early detection to prevent excessive expenditure when the disease spreads broadly. These fundamental approaches can be implemented in agricultural, biological, pharmacological, and cultural practices to control plant diseases.

This study avails the following contributions presented:

- A novel IX-CNN model is proposed considering depth-separable and inception convolution layers to circumvent the overfitting issue and improve the model performance by elevating the inadequate feature maps of the inception convolution layer.
- Design and develop an end-to-end approach using IX-CNN, transferring learning to identify affected crops using popular datasets.
- To test the proposed model's performance, we considered evaluation parameters such as model accuracy, precision, sensitivity, recall, F1-scores, and confusion matrices. Popular classifiers, including random forest (RF), support vector machines (SVMs), decision trees (DTs), and K-nearest neighbours (k-NN), were also used.
- The proposed IX-CNN model demonstrated an accuracy of 100% on Plant Doc, PlantVillage, and Turkey Disease datasets. It also showed an accuracy of 99.79%, 99.95%, and 98.64% on Rice Disease, RoCole, and NLB, respectively.
- Using the MATLAB deep learning toolbox, Android and servlet models operate in real-time, aiding farmers who would have used traditional approaches, such as naked eyes or experience, to detect and classify diseases.

The remainder of this study is structured as follows. Sect "Related works" presents related work entailing the application of deep learning in enhancing agricultural sustainability. Sect "Theoretical background" discusses the methodology and system design concerning the main contents: architectural design, the user (application user) design of the interface, and principal component design. Sect "Methodology" presents the proposed system implementations for feature extraction, segmentation, learning, and classification, as well as the computer-based format of the proposed model and an Android application. In Sect "Results and discussions", results and discussion are demonstrated, where evaluation and testing are done in three ways: segmentation, testing, and evaluating different classification classifiers and edge-cutting models. Finally, in Sect "Conclusion", model limitations and conclusions are presented, and future work is illustrated.

Related works

Recent studies demonstrate that the application of ensemble-based strategies on the existing pre-trained network ResNet, AlexNet, and VGGNet reduces infectious *Elettaria cardamomum* (cardamom) disease misclassification that is imperative to human life; for instance, the multilevel feature fusion network (MFFN) incorporated adaptive attention mechanisms, enhancing robustness by considering diverse network features¹⁴. A systematic review of 160 diverse machine learning (<https://www.ibm.com/topics/machine-learning>) and deep learning (<https://www.ibm.com/topics/deep-learning>) approaches for automated plant disease detection highlighted cost-effective, scalable solutions to reduce dependency on domain experts. The study presented 50 datasets, including real-time and hybrid models; it emphasised hyperparameter significance and identified key challenges in plant disease detection, such as data quality and availability, real-world variability, computational demands, scalability and cost, among others¹⁵. Furthermore, with the integration of ResNet50 with adaptive

attention mechanisms (channel, spatial, and pixel attention), the model boosts feature extraction, even with challenging dataset traits such as data imbalance and similar features among classes, as earlier demonstrated¹⁶. When it comes to noise handling involving Aroche methods, the performance differences, results quality, and noise ratios often hinge on the specific Aroche algorithm variant used and its application context¹⁷. Within this research, an adaptive version incorporates adaptive filtering, which can dynamically adjust to varying noise levels. This often yields a lower noise ratio, enhancing image clarity. When it comes to execution speed and efficiency, it repeatedly requires fewer computational resources, making it suitable for real-time or near-real-time applications, but it might miss present details in complex scenes, as illustrated¹⁸. The adaptive Aroche variant usually performs better in challenging environments, sometimes achieving over 96% or higher accuracy due to dynamic adaptability as the results as presented.

Plant disease region segmentation enhancement was presented using deep learning segmentation architecture (UNet) through the integration of dilated convolution and was tested on the bacterial leaf blight, brown spot, and leaf smut of the Rice Leaf dataset as this research used during its training³³. For each disease, authors created manual ground truth mask images. A comparative analysis reveals that the modified architectures outperform their unmodified versions. The enhanced UNet model demonstrates a mean loss of 0.3018 and a mean Dice coefficient of 0.6785, representing a substantial improvement over the conventional UNet model¹⁹. Timely detection of rice crop conditions is crucial for effective damage control, motivating extensive research in rice monitoring using advanced machine learning and deep learning techniques. A study examined key research trends, data types, these models, pre-processing, segmentation, and feature selection in rice health monitoring. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (<https://www.prisma-statement.org/>) methodology, where seven critical research questions were addressed to guide future research in rice disease and disorder detection²⁰. In another study, the authors leveraged a transfer learning (<https://www.ibm.com/topics/transfer-learning>) approach on the pre-trained architectures DenseNet201, InceptionResNetV2, MobileNetV2, ResNet152V2, InceptionV3, Xception, and VGG19 for diagnosing rice diseases across two publicly available datasets. To optimise diagnostic accuracy, an ensemble approach combining various transfer learning models was implemented and rigorously tested. Results indicated that most ensemble configurations outperform individual models, with certain ensembles demonstrating superior accuracy even compared to advanced frameworks, for instance, Convolution-XGBoost²¹. Evaluating several unsupervised machine learning techniques, K-means, FCM, SOM, mean-shift, GMM, and HMM for classifying Sentinel-2 data was proposed. Mean shift efficiently classified continuous vegetation in Study Area 2 but confused vegetation and fallow land in Study Area 1. Similarly, SOM accurately identified built-up areas in Study Area 2 but had notable misclassifications in Study Area 1, as depicted²².

Theoretical background

DL models have multiple layers, each with an explicit data analysis and interpretation purpose. These layers can be sketchily categorised into two categories: convolutional and fully connected.

Background

This section details the two main blocks and their functions within a DL model. Another technique that can be applied to convolutional layers is padding. The padding adds pixels to an image's edges before a CNN kernel processes it. This is done to provide the kernel with additional space to process the image, which can assist in extracting low-level features and improve the accuracy of image analysis. Padding can be added to the top, bottom, and left, and the user can specify the right of the image and the number of pixels added.

Inception layer

The convolutional layer, the Inception Layer (<https://www.analyticsvidhya.com/blog/2018/10/understanding-inception-network-from-scratch/>), was first used in the film Inception. Google created an architecture to help with image classification tasks. The inception layer extracts the feature at various scales from the input data (images), which can reduce the computation necessary while improving the model performance²³. The Inception layer comprises the input data through various branches, each using a different set of filters, with each branch having a different kernel size. As a result, the Inception layer can extract features from the data. Input data at various scales, obtaining more details, and enhancing the model's efficiency. Each branch's output is combined and passed on to the following layer²⁴. The Inception layers are created to be computationally efficient with the most significant feature of this technology. Efficient and capable of multiple scales of feature extraction from the input data while maintaining that the model only has fewer parameters²⁵. The Inception layer architecture was created to reduce the increasing depth and breadth of neural networks and several parameters (as Fig. 1 demonstrates). Layers of the inception modules are constructed using 1×1 , 5×5 convolution, and max pooling of 3×3 . The architecture that results from these modules being stacked on top of one another is what the Inception layer is composed of. The Inception layer generally seeks to learn distinct features by reducing the input's dimension and modifying it with different convolutional techniques, concatenating the results of the filter sets together.

Depth-separable convolutional layer

The parameters and computational complexity of depth-separable CNN layers are less complicated than those of ordinary convolutional layers. A single convolutional filter individually processes each input channel, and every input channel gets the feature map's output. The depthwise convolution in the pointwise convolution stage of a 1×1 convolution is used to join the output feature maps²⁶. Condensing CNN's parameters improved object recognition and image classification due to computational complexity recognition. Additionally, it is well known for creating CNN architectures for embedded and mobile devices with constrained processing power.

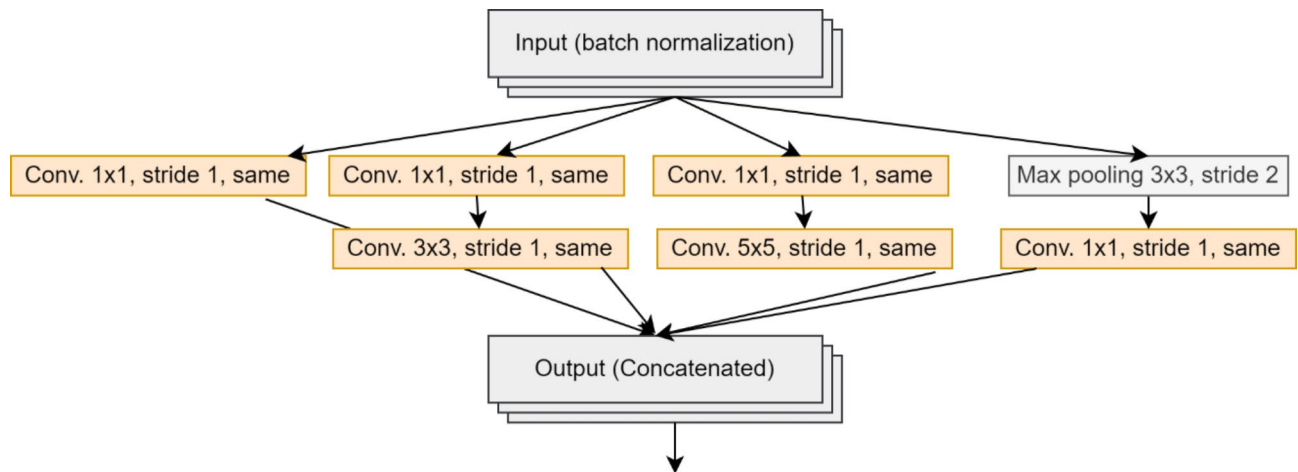


Fig. 1. Structure of the inception layer.

Performance evaluation parameters

This subsection briefly overviews the model performance evaluation parameters employed in the research experiments.

Performance evaluation parameters

Standard classifiers, including LR^{27,28}, SVM²⁹, and R¹¹, are often used to evaluate these pretrained CNN networks in deep-learning methodologies. Table 1 illustrates the testing accuracy for every class using different classifiers. Additionally, the accuracy, recall, and precision of the model are evaluated using true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN)^{12,13}. TP shows instances where the model accurately identified them as positive. TN displays the instances that were accurately classified as negative. In those instances, the model accurately predicts the negative class; therefore, it corresponds to those instances²⁷. The number of times that the model mispredicted as positive is summed up by the acronym FP. FN stands for a situation where the model mistakenly projected a negative outcome.

Accuracy Equation (1) expresses the ratio of correctly completed predictions to all data points collected (T).

$$\text{Accuracy} = (\text{TN} + \text{TP}) / T \quad (1)$$

Precision Equation (2) shows the percentage of samples that were positive to the total samples that were expected to be positive.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (2)$$

Recall and sensitivity The terms “recall” or “sensitivity” are defined as the proportion of correctly anticipated positives to the total number of actual positive results; Eq. (3).

$$\text{Recall} = \text{TP} / (\text{FN} + \text{TP}) \quad (3)$$

F1- scores This refers to the harmonic mean of precision and sensitivity (recall) calculated in Eq. (4).

$$\text{F1 - Score} = 2 * (\text{Recall} * \text{Precision}) / (\text{Recall} + \text{Precision}) \quad (4)$$

Classifiers

During the study, we considered several classifiers, for instance, artificial neural network (ANN)³⁰, RF^{31,32}, DT^{33,34}, SVM³⁵, and subjected it to the used datasets.

Datasets

This study uses six different datasets to train and test the proposed IX-CNN model. All considered datasets were marked manually by different institutions’ scientists. The datasets were categorised into three parts: training, validation, and testing. 70% were used for training, 15% were used for validation and 15% for testing (details on how each dataset was divided are presented in supplementary material, tabular form, and sampled image illustrations).

Plant Village dataset

The famous PlantVillage dataset is made up of 54,303 images of 38 different plant species’ healthy and unhealthy leaves. Each of the 38 categories in the images corresponds to a different plant type and illness. To illustrate

this, the category “Tomato_Bacterial_spot” contains images of diseased and healthy tomato leaves infected with bacterial spots³⁶. The dataset was created under the PlantVillage project, a collaboration between researchers at the University of California, Davis, and the University of Wisconsin-Madison. It is considered the most valuable resource for researchers and developers on plant disease detection and diagnosis³⁹. The dataset can be used to train and evaluate machine learning models for PDDC. We use the reported configuration during this research to produce the best detection and classification performances²⁸.

Turkey Plant Disease dataset

This is one of the newest Apple disease datasets from Turkey’s Bingol and Malatya cities. It was attained using A 7200d smart camera, with effective pixels (megapixels) of 24.2 million, sensitivity to – 3 EV, ISO 100, RGB metering sensor, with a 2016-pixel 1.2 M dot RGBW LCD, amplified depth of buffer, and other specifications³¹.

Plant Doc dataset

PlantDoc dataset, which consists of 2,569 images from 13 plant species and 30 classes (healthy and diseased). Researchers at the Indian Institute of Technology, Delhi, created the dataset. PlantDoc dataset can also be used to develop mobile apps that farmers can use to diagnose plant diseases even though it has limited image counts⁴⁰.

Rice Disease dataset

The Jiangsu Academy of Agricultural Sciences Institute of Plant Protection in Nanjing, China, provided 4684 rice leaf images, where 1604 leaf images are bacterial blight (BB), 1620 brown spot (BS), and 1460 leaf smut (LS) images. Plant protection experts examined and confirmed the images to ensure quality. High-resolution images of rice leaves exhibiting symptoms of the specified disease. This aids in visual diagnosis and machine learning-based image recognition tasks. Textual description outlining the major symptoms visible on the leaf, offering a more detailed understanding of the disease’s progression and manifestation⁴².

RoCole dataset

This dataset contains 1560 images of robusta coffee leaves. With visible mites and spots (indicating the presence of coffee leaf rust) for infected cases and images devoid of these structures for uninfected cases⁴³. The data set contains descriptions of the state (unhealthy and healthy) and the severity of the disease (leaf area with spots). The photographs were taken in Ecuador’s robusta coffee plantations. They were captured with a smartphone camera at a working distance of 200–300 mm without the use of magnification⁴⁴.

NLB dataset

The NLB plant dataset is images of maize (corn) leaves annotated with northern leaf blight (NLB) lesions. The dataset comprises three image sets with the corresponding annotations (Handheld with 1787, 7669, Boom with 8766, 55,919, and drone 7669, 42,117 images and annotations, respectively). All photos were captured during field tests of maize that had been spread with *Setosphaeria Turcica*, the disease-causing agent⁴⁵. The NLB plant dataset contains 18,222 images of maize leaves, of which 105,705 are annotated with NLB lesions⁴⁶. The images were collected from maize fields in California using various methods, including hand-held cameras, cameras mounted on a boom, and cameras mounted on a UAS. On the NLB, the data had some limitations, including the extent and margins of lesion axes are not specified in their annotations. The credibility of annotations cannot be determined by any means. Others are partially hidden, blurry in the dark, or overpowered by the sun³⁰. Experts could find it challenging to recognise NLB from diseases like Stewart’s wilt or anthracnose leaf blight from a distance. On-foot phenotyping without similar disorders does not rule out the possibility of false-positive outcomes. All images were taken at a field in New York State. Due to the possibility of different disease symptoms in different places, this limits the generalizability of the findings.

Methodology

Inception and depth-separable hybrid architecture for a CNN convolution layer are presented. In comparison, the depth-wise separable convolutions have two advantages, like the conventional convolutions. First, they lessen the problem of overfitting because of the reduction in the number of parameters. Second, compared with other methods, its computational complexities are the current model. This makes a promising method for mobile devices called depth-wise separable convolution applications for vision. The performance of the model could nevertheless be affected by the parameter reduction. On the other hand, conventional convolutions result in costly computation and overfitting issues after prolonged use.

Considering these two motivating details, we propose a new model consisting of layers of inception and depth-wise separable convolution that are both attained in the computation. We use depth-wise separable convolutions and enrich them with the Inception convolution layer and restricted feature maps. The inception layer proposes that, as an alternative (Fig. 2), the network can grow wider instead of deeper by adding more layers, and one can extract more features. Additionally, the depth-wise separable convolutions lower the model’s parameters and overfitting. The proposed model is subdivided into three structural phases: namely, Entry, Middle, and Exit. Firstly, the data passages via the entry phase, then two times through the middle phase, and finally through the exit phase. Convolution is performed on the input in the inception layer using three filters of varying sizes (5×5 , 3×3 , and 1×1). Then, 1×1 convolution and max-pooling are applied. Prior to sending the outputs to the following layer, they are combined. The max-pooling layer follows this step to reduce the number of features. The depth multiplier used by all Separable Convolution layers is 1 (no depth expansion).

Each separable convolution layer then batch normalisation. The model’s output layers are dense layers with N nodes that are based on the classes in the dataset, and in this layer, softmax has been used as a classification function. In the entry phase, there are three inception layers, and their filter depth is increased gradually twice;

thus, the 1×1 filter of the first inception layer is 64, while in the second, together with the third, it is 128 and 256, correspondingly. The same is true for the 3×3 and 5×5 filters, where their depth in the first inception is 128 and 32, respectively. The inception layer in the middle and exit phases have a constant depth size for their filters. The depth of the 1×1 , 3×3 , and 5×5 filters are 256, 512, and 128, respectively. All sizes were chosen carefully to match the residual layers added from a while to a while, as shown in the model structure flow graph.

The middle phase of the Xception model is repeated twice, which causes a large increase in the model's parameters. Increasing the parameters in a DL model can have several effects on the model's performance.

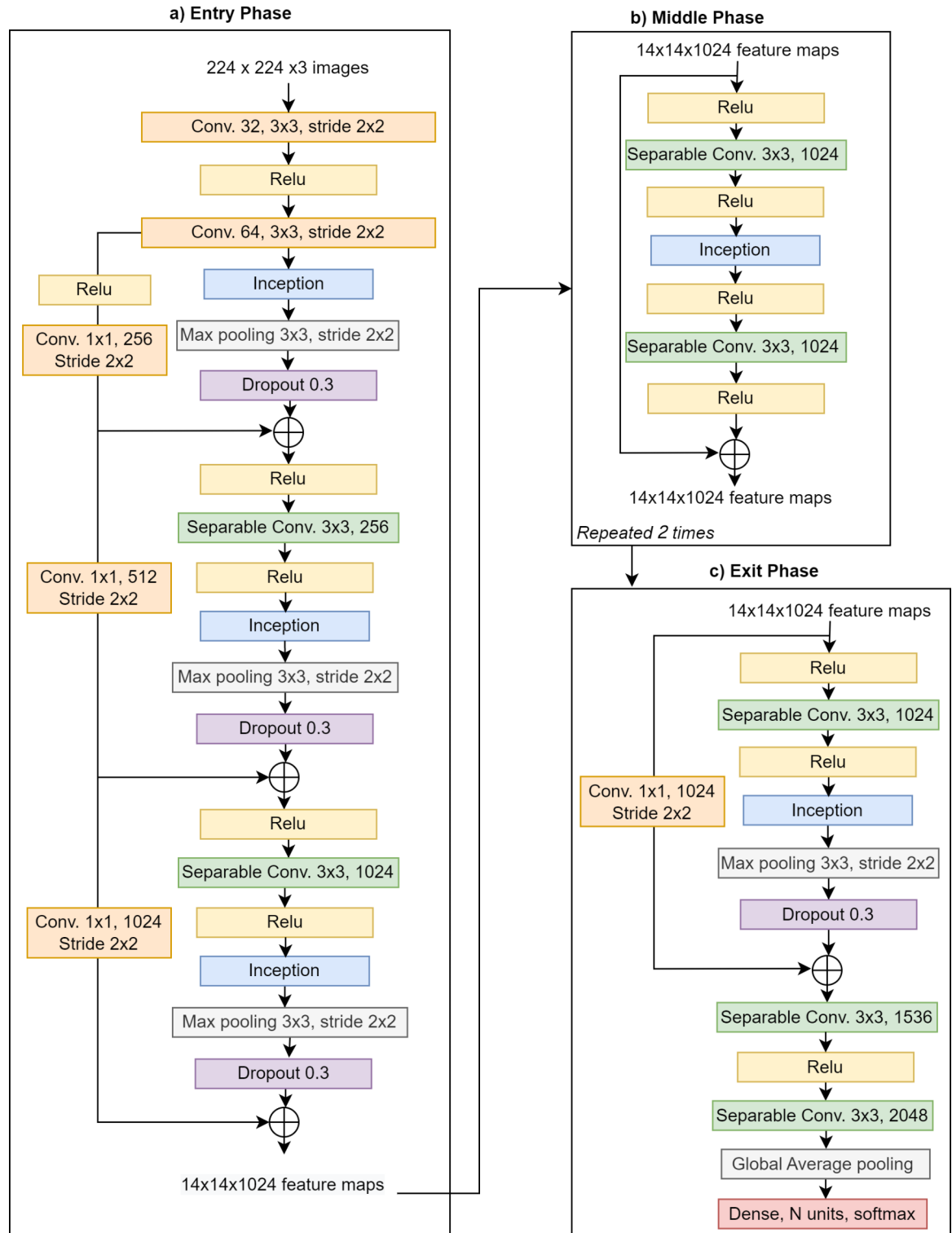


Fig. 2. Proposed structure of IX-CNN model.

In general, increasing parameters can lead to greater robustness, meaning it can fit the training data better and achieve higher accuracy on the training set. However, accumulating increased parameters increases the overfitting risk when the model does well in the testing phase but terribly on unknown data. This is because the proposed method with more parameters is more flexible and can more easily learn patterns in the training data that do not rely on new data generalisation. Regularisation methods, such as weight decay or dropout, are often essential to mitigate overfitting risks. Adding more parameters may make the model more computationally demanding, needing more memory and computing capacity to train and evaluate. To obtain a better trade-off between model capacity and generalisation performance, it is crucial to balance the parameters in a DL model carefully. To reduce the number of parameters in the suggested model, we advise repeating the middle phase only twice rather than eight times.

Methods of architectural design and development

The system architecture demonstrates how the application works and interacts with users. In the “main development component design” document that will be made, the development plan for each principal component will be described in detail. Figure 3 illustrates the general organisation of the architectural implementation design. The first component is the Android app, which acts as a client and provides an interface to the user (in this regard, the farmers’ application interface) through which users can interact with the application’s capabilities. The user can take a picture using the mobile phone’s camera or select an image from the phone’s image collection to serve as the image of the leaf to be diagnosed. When the user has selected the photo from the photo gallery of a smartphone or decides to take a picture of a fresh plant leaf, the Android application will be used to submit that considered photo for processing.

They were followed by the next section, where a server written in Java servlets is developed. Its task is to make connecting the client and the primary services easier. The servlet will first create a socket with one part for each new connection. It will then receive a photo to execute the primary function process of disease processing to detect whether the plant photo has disease or is disease-free and send the output to the client. The MATLAB model oversees the putting together of the third part of the app, which is essential because it is used to find out what kind of disease it is. It is also the most challenging and vital section of the program.

Main component design

The section details the significant design development and the proposed model plan that will be implemented on the server side. Since the MATLAB-based recognition program is the most crucial part of finding a solution to the PDDC problem, it is essential to show how the program is made to be used.

Phase I of main design

During this phase, a few steps must be taken for the MATLAB code for plant disease detection to work.

- Image acquisition: An image is acquired from the side of a smart gadget, such as a smartphone or a system (computer).
- Image pixel reduction: Since the image captured or chosen from the phone or computer could have a size, at this stage, the image is shrunk to a size that is good for the user in terms of how much time it takes and how well it is recognised.
- Leaf segmentation: Segmenting an image is breaking down a digital picture into smaller parts called objects or regions. The segmentation target is to make an image’s representation less complicated so that it can be better understood and analysed. In this research, we consider the EM algorithm to ensure proper segmentation is done.
- Post IP: Relevant features of the segmented image are preserved, like yellow and green features, leading to removing small unnecessary regions or objects. Lastly, filling the removed small leaf holes is necessary to attain a whole plant leaf.
- Disease segmentation: segment the plant leaf detection with several segmentation approaches, compare these methods, and use the one with the best accuracy for disease detection.
- Feature extraction: Different features are extracted from the segmented images, as demonstrated in post-IP, that can be used to classify various leaf diseases.
- Feature matrix generation: This is the last stage of Phase I, where a combination of different descriptors is turned into a training feature matrix.

Phase II of main design

This is the second phase of the component design and development. It involves different stages, as precisely explained below.

- Feature matrix training: Following step I, when the other descriptors are combined to train the feature matrix, at this level, we use different training methods to prepare the feature matrix for both the whole leaf features and disease features.
- Evaluation: The model is evaluated by assessing the performance of the considered approach on the entire plant leaf and disease features. Moreover, compare and select the best one for disease detection, identification, and classification.
- Recognition: For every image presented to the model, we use the previous methods to create the distinguishing feature vector. The trained classifier then classifies which category the feature belongs to.
- Application development: After implementing all the phases in MATLAB, an Android version of this model is developed and implemented like in a GUI, adding a function that enables the user (the farmer) to select from

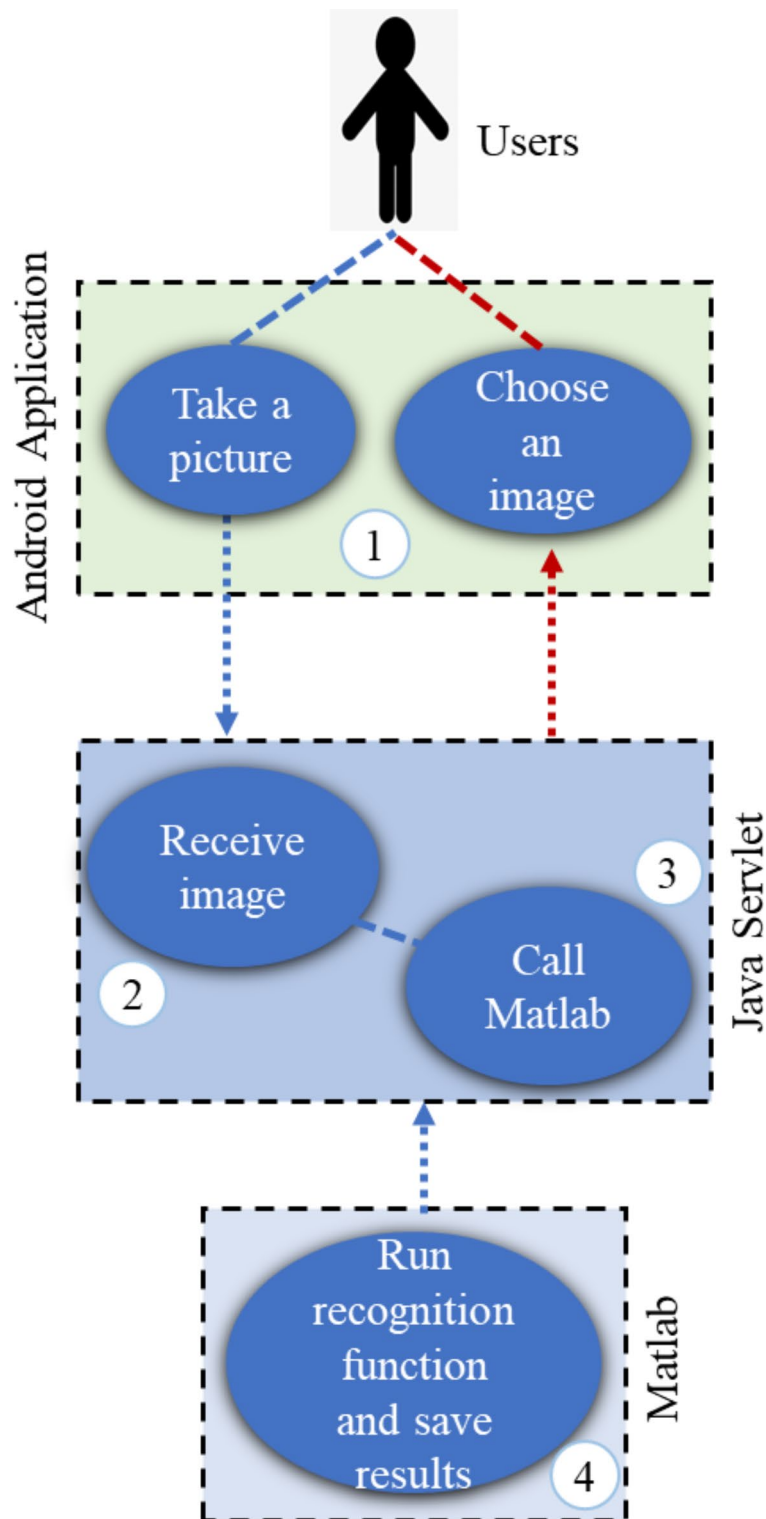


Fig. 3. User system communication.

the phone's gallery or take a picture of the plant directly. Moreover, a function that allows the application to send the images to the server should be added.

- **Server development:** Develop a server-side program using a servlet that receives images from the client and enables them to be called in a MATLAB-coded program.

Model implementation

This section details how the model was implemented, as seen through the research's complexity.

Learning and classification

The system was split into three distinct parts established on the mode of operation. This would make the implementation, verification, and documentation processes easier to handle. The first stage is the Android app, which acts as a client and allows users to interact with the app's features through a user interface. The user can either take a picture of the leaf with the phone's camera or choose an image from the phone's image library to use as the image of the leaf. When the user selects the picture of a leaf they want to use, the developed AI app will send that picture to the next section. The second stage is a server written in Java servlet. Its task in this regard is to make it easier for the client and the primary services to communicate. After getting an image as input, the servlet is programmed in such a way that it opens a socket with a port for every connection. In the last stage, the main component task is IP and detecting whether the image plant is affected, then sending the output image results back to the client. MATLAB oversees putting together the third part of the app, which is the most critical part because it is used to identify what kind of disease it is. It is also the most challenging and vital section of the program. The following part will provide a high-level overview of the diagnostic program design. After the recognition process is completed, the result will be saved on the server in preparation for transmission to the clients in acceptable computation time.

Reduction of dimensions

Correlation feature selection (CFS) is an important aspect that seeks to determine which features should be selected to build a classification model for a specific concern using a correlation-based approach. A high-quality feature subset can increase the interpretability of the model, cut down on the time spent in training, and boost generalisation by lowering the likelihood of overfitting. The idea is that a good feature set should have a high correlation with the class in question while having little to no correlation with other classes. CFS is an algorithm that combines a search strategy, the Pearson correlation measure, and an evaluation of the heuristic merit function that has been given. The project uses forward selection to find appropriate feature subsets for dimension reduction. Then, we generate a Pearson correlation matrix (PCM) between each pair of features and classes and an empty feature index set. Then, we constantly add a new feature to the specified feature set to make the value of the new feature subset's merit function the largest until the merit function converges. At this point, we cannot add a new feature. The coefficient demonstrates how far a data point is off the line that a PCM draws as the best fit through the data. The ratio of X and Y's covariance and standard deviation yields the Pearson correlation coefficients.

Results and discussions

This section presents the experimental results and corresponding discussions of the IX-CNN model and its deployment.

Generalization and overfitting

Overfitting is a complex training algorithm issue. When the training times rise, the training set error will decline. However, validation and test set errors are likely to increase. The classifier may lose generalisation to new data. Early stopping prevents overfitting. The MATLAB deep learning network splits data into train, test, and validation sets. The training set updates the bias, weights, and network gradient. Validation subsets monitor training. Training reduces validation errors. Validation may rise with each iteration, causing overfit. As the validation error increases, the training process stops and returns the minimum error. MATLAB networks start with distinct weights and biases, resulting in different training, test, and validation sets. To guarantee perfect neutrality, we split the original data into two groups: one for network design (2/3) and one for testing (1/3). Determine the average mean square error. The average performance will compare the network efficiency and accuracy for each topology and parameter setting.

Performance evaluation of the proposed IX-CNN model

The experiments were conducted using six different datasets, and the highest accuracy score was attained, as demonstrated in Table 1, using pretrained CNN model results. Figure 4 illustrates the accuracy assessment proposed IX-CNN in relation to pre-trained CNN models based on the different datasets; for instance, the proposed model attained 100% when tested on the Plant Doc, PlantVillage, and Turkey Disease.

Performance evaluation of the machine learning algorithms

The efficiency of retrieving data can be evaluated using several methods, including recall and precision. Calculating the number of positives retrieved and subtracting it from the overall number of positives, including those that were not recovered, is how the recall method describes how thorough the process was. The term "precision" refers to the accuracy of the retrieval, which is determined by calculating the number of correctly retrieved positive data and dividing that number by the number of all correctly retrieved data, which includes the number of correctly retrieved harmful data. In other words, precision describes the accuracy of the retrieval. These can be utilised to evaluate the effectiveness of the classifier in question. Although many metrics can be used for classifying data, this study aims to use the F1-score, precision, and recall in relation to the selected classifiers of machine learning algorithms and the datasets to explain the circumstances in which its application is beneficial in disease spot classification, as expressed in Table 2.

Figure 5 presents different performances when RF when subjected to the used datasets; it attained the best F1-score at 98.22%, 96.24%, and 96.035% on PlantVillage, PlantDoc, Turkey Disease, Rice Disease, and RoCole datasets. NLB demonstrated at 86.96% on F1-Score. More still, it attained 100% on Turkey Disease on precision, 97.14% on PlantDoc, and 96.75% and 100% on the PlantVillage on Precision and Recall, respectively. When the datasets were subjected to the SVM, the best F1-Score achieved was 95.13%, 94.11%, 93.17%, 93.96%, 93.89%,

	Dataset accuracy (%)					
	Plant Doc	PlantVillage	Turkey Disease	Rice Disease	RoCole	NLB
NASNet	79.75	82.68	83.48	84.38	89.68	86.18
EfficientNet	89.98	83.25	89.61	89.87	83.89	86.87
ConvNet	89.91	82.90	84.23	81.84	70.89	90.40
IX-CNN (proposed)	100	100	100	99.75	99.95	98.64

Table 1. Accuracy evaluation of the proposed IX-CNN and pre-trained CNN models.

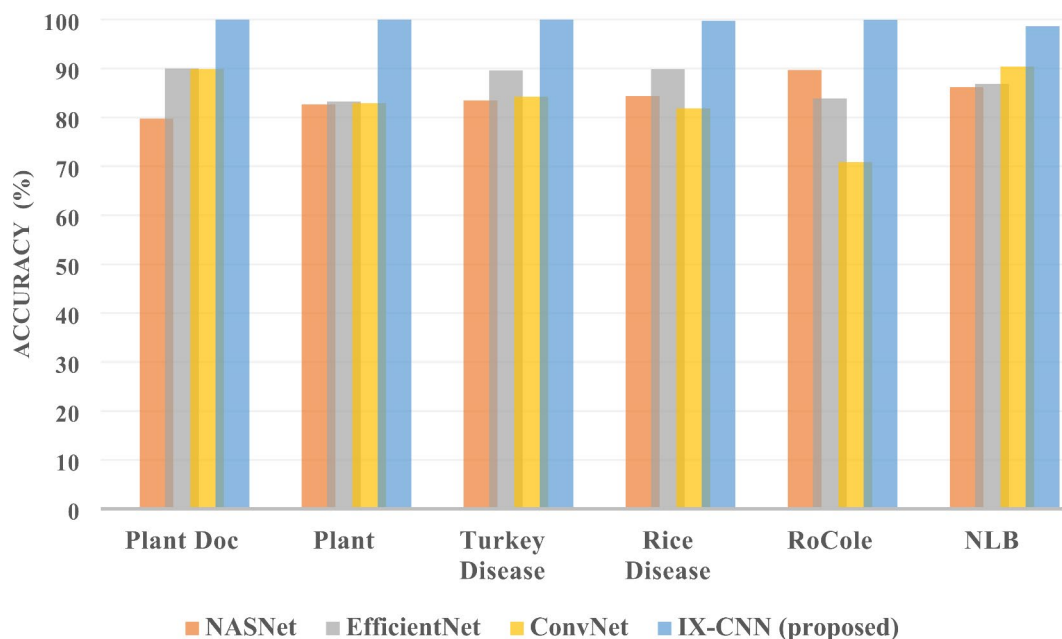


Fig. 4. Proposed IX-CNN and pre-trained CNN models accuracy assessment based on the different datasets.

and 82.86% on PlantVillage, Turkey Disease, PlantDoc, Rice Disease, RoCole, and NLB, correspondingly. On Precision, PlantVillage attained 93.28% and 97.50% on recall, as illustrated in Fig. 6. Figure 7 demonstrates that the Plant Doc dataset attained the best F1-Score of 97.09% amongst other datasets, with a Precision of 97.50% on the PlantVillage dataset and Recall at 98% on the Plant Doc dataset. Amongst all machine learning algorithms, DT had generally lower performance in relation to the other classifiers; it was able to attain 94.82%, 91.72%, and 98.75 on F1-Score, Precision, and Recall, correspondingly. The lowest F1-Score was attributed to DT at 77.75%, as illustrated in Fig. 8. In terms of F1-Score performance, RF outperforms SVM, ANN, and DT when subjected to the classification problem, as it obtained 98.22%, 95.13%, 96.83%, and 94.82%, as presented in Table 2.

Comparison with the pre-trained models

To further assess the proposed model, many techniques in literature have proven effective for accurately categorising data in various fields. However, our proposed IX-CNN model is straightforward and innovative, making it easier to implement than advanced approaches. Experimental results using the suggested methodology were contrasted with those obtained using other methods described in the literature. The presented model attained higher accuracy than other counterparts, such as^{26,28,29,31,35,41,42}, using considered and accessible datasets, with values higher than 98%. This suggests that the proposed method is a successful and practical approach to solving the problem, and an analysis of available applications and diagnostics is explored in Table 3.

Model deployment

The proposed model demonstrated exceptional performance, as detailed in Table 3, for identifying several plant leaf diseases when tested on the six popular datasets entailing different crops, offering numerous benefits, including the method showing high effectiveness in accurately classifying diseases and enabling early intervention to prevent crop loss and yield reduction. The model effectively handled data imbalance and differentiated between leaf textures across various plant diseases, depicting robust feature extraction. Nevertheless, the method can be adaptable for various crops beyond those that are within the datasets, allowing for a broader application in agriculture. The practical implementation of the crop classification system in the field involves the following:

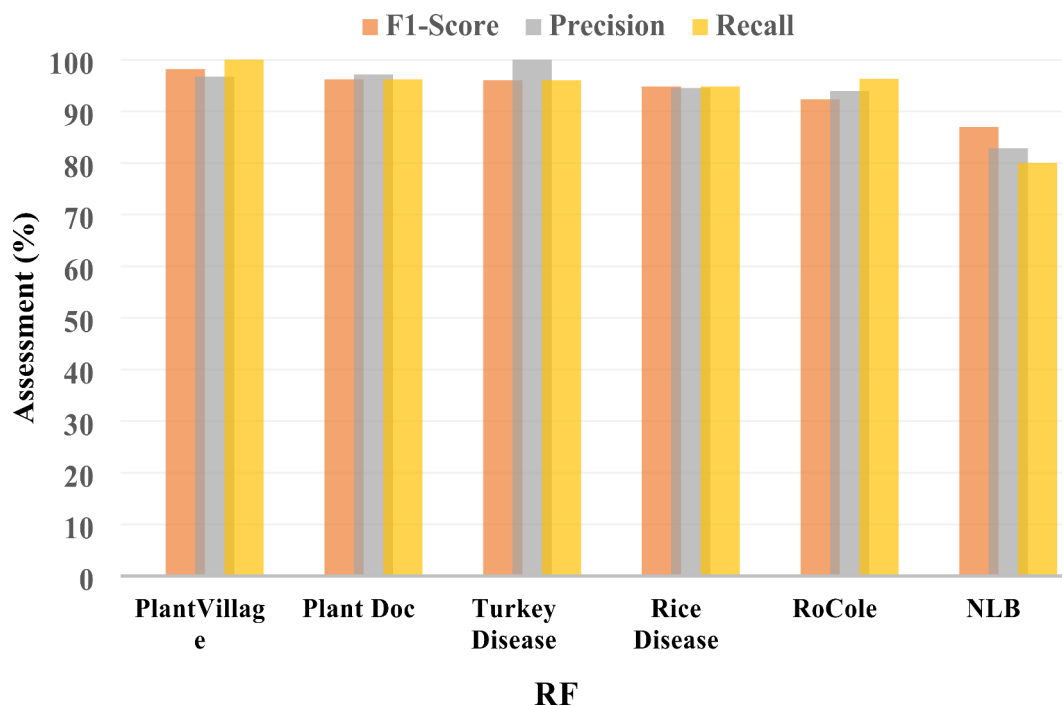


Fig. 5. Performance assessment on the random forest.

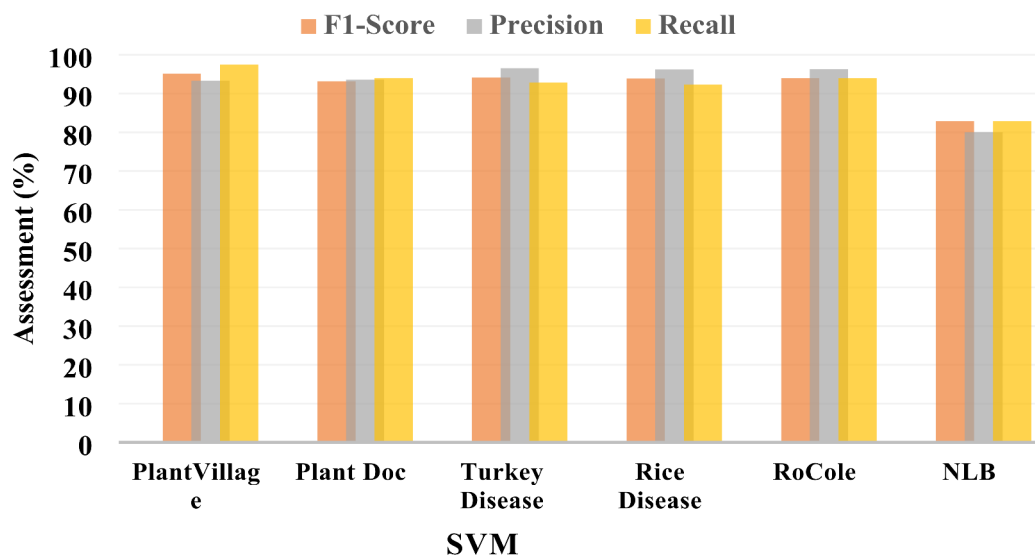


Fig. 6. Performance assessment on the support vector machine.

- (i) primary users (farmers or agricultural extension workers, agricultural workers and agronomists and researchers) using smartphones equipped with cameras to capture high-quality images of plant leaves.
- (ii) The captured images can be transmitted wirelessly to a local processing unit via Wi-Fi or mobile data, depending on the internet connectivity.
- (iii) The local processing unit can run the trained proposed model to classify the images in real-time, providing instant feedback on the health status of the plant diseases or pests and recommended interventions. The identified pest or disease is then displayed on the screen, presenting the identified disease and pest name, and suggesting the appropriate pesticide; the system presents the option to the use explore more about the disease or pest identified.

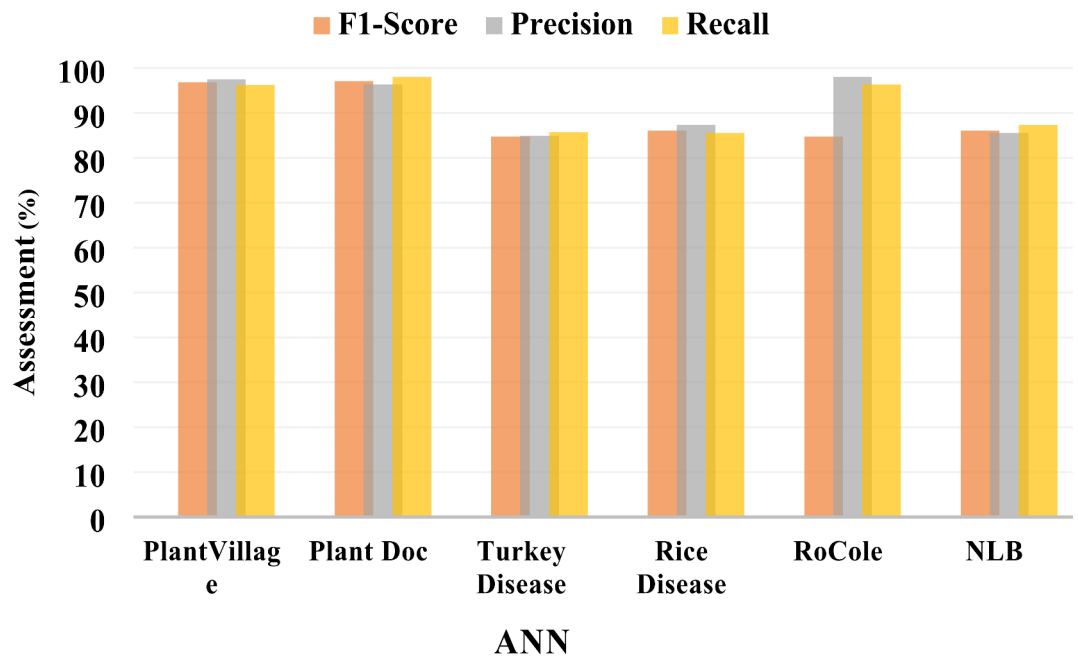


Fig. 7. Performance assessment on the artificial neural network.

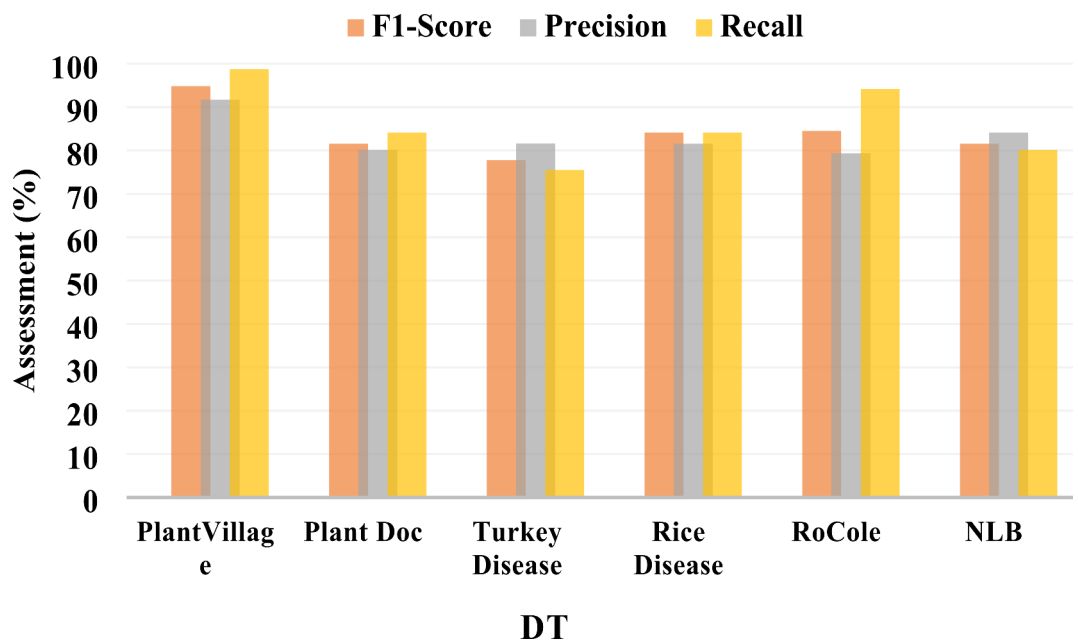


Fig. 8. Performance assessment on decision tree.

- (iv) A user-friendly mobile application can be developed, allowing users to upload images easily, receive classification results, and access treatment recommendations as demonstrated within the supplementary file.

This research can be enhanced and sustained by improving different geographical responses to the farmer, making it more interactive, and providing more details about the plant's disease cure and how to use the pesticide. Other various categorisation algorithms can improve this project as well as several studies have demonstrated that the integration of mobile applications and deep learning for plant disease management³⁷, emphasising their potential for real-time diagnostics in the field study, explores the use of smartphone apps for crop monitoring, demonstrating improved yield management and disease detection through immediate feedback³⁸. In the future, we intend to make avail a mobile application that can be installed on Windows, iOS, and Android phones so that users can use it anytime. The current model size is quite large and may not be ideal to be installed on low-

Learning methods	Assessment criteria	Datasets					
		PlantVillage	Plant Doc	Turkey Disease	Rice Disease	RoCole	NLB
RF	F1-Score	98.22	96.24	96.03	94.84	92.33	86.96
	Precision	96.75	97.14	100	94.53	93.96	82.86
	Recall	100	96.24	96.03	94.84	96.33	80.00
SVM	F1-Score	95.13	93.17	94.11	93.89	93.96	82.86
	Precision	93.28	93.57	96.53	96.21	96.33	80.00
	Recall	97.50	94.00	92.86	92.32	93.96	82.86
ANN	F1-Score	96.83	97.09	84.72	86.07	84.72	86.07
	Precision	97.50	96.33	84.88	87.32	98.00	85.54
	Recall	96.25	98.00	85.71	85.54	96.33	87.32
DT	F1-Score	94.82	81.54	77.75	84.11	84.49	81.54
	Precision	91.72	80.10	81.61	81.54	79.33	84.11
	Recall	98.75	84.11	75.48	84.11	94.17	80.10

Table 2. Performance evaluations on different classifiers tested on the datasets.

Reference	Accuracy (%)	Dataset
28	83.06	Real-field images
31	91.2	PlantVillage
29	93.75	PlantVillage
35	95	ALE
33	94.64	Plant village
42	86.01	UCI machine learning repository
36	89.19	Tomato powdery mildew
38	92.15	Real-field images
Proposed IX-CNN	100	Plant Doc
	100	PlantVillage
	100	Turkey Disease
	99.79	Rice Disease
	99.95	RoCole
	98.64	NLB

Table 3. Comparison with the currently proposed models and commercially available approaches.

memory end-user devices^{47,48}. However, it has been revealed that applying quantisation to any designed CNN model reduces the model size without affecting its performance, which can be another innovation.

Conclusion

This research aims to help traditional farmers identify the leaf disease affecting farmers’ farms based on the leaf appearance. It has the potential to serve as a central prototype that can be utilised by the agriculture sector (in case a computer-based system is used) to enhance the production of crops or by gardeners to identify the disease (using the availed app), supporting them to stop disease spread on the farm. The segmentation algorithm segments leave illnesses without fault. Extracted feature-based machine learning works well. Cross-validation tested the learning method’s generalisation. Both disease spot and leaf features are trained by employing the most prevalent machine learning applications, including SVM, ANNs, DTs, and LR. IPM, PlantVillage, and Turkey Disease datasets demonstrated an accuracy of 100%, yet Plant Doc, Rice Disease, RoCole, and NLB attained an accuracy of 99.79%, 99.95%, 98.64%, and 96.46%, respectively. The results show that the features collected and the classification approach suit this application. More training samples will boost the result. The app is simple to use. In the future, more experiments will be conducted on other plant leaf diseases besides the three plant leaves that affect the global food security of leaves to broaden the scope of application so that it can be used on as many kinds of plants as feasible. The application program can incorporate additional features and capabilities, such as viewing histories and saving locations.

Data availability

All data generated or analysed during this study are included in this published article [and its supplementary information files].

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Author contributions

W.S. was responsible for conceptualization, methodology, investigation (lead), data curation, formal analysis (lead), writing the original draft, and software (lead). A.T. was responsible for funding acquisition, methodology, resources, supervision, visualization, and reviewing and editing. C.L.D. was responsible for conceptualization (supporting), supervision, reviewing, and editing (equal). R. A. A. H. M. was responsible for supervision, conceptualization (supporting), and reviewing and editing (equal).

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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